Basic Economic Concepts

All 36 topics. Read the rule, inspect the graph, then recall it without looking.

1.1 Scarcity & opportunity cost

Limited resources force choices. Factors: land, labor, physical capital and entrepreneurship. Opportunity cost is the value of the **next-best alternative**. Positive statements are testable; normative statements express value judgments.

1.2 Allocation & economic systems

Every economy chooses what, how and for whom to produce. Markets use prices; command systems use central decisions; mixed systems use both. Households sell factors and buy goods; firms do the reverse.

1.3 Production possibilities curve (PPC)

On the curve: productively efficient. Inside: underused resources. Outside: currently unattainable. Bowed-out PPC: increasing opportunity cost. More resources or better technology shift PPC outward; lower unemployment moves the production point toward it.

> Opportunity cost of X = Y forgone / X gained.

1.4 Comparative advantage & trade

Absolute advantage: more output per input. Comparative advantage: lower opportunity cost. Specialize by comparative advantage. Mutually beneficial trade terms lie between the two opportunity costs. Trade can expand consumption beyond a country's PPC, not its production.

> Output table: OC of X = Y output / X output. Time table: OC of X = time for X / time for Y.

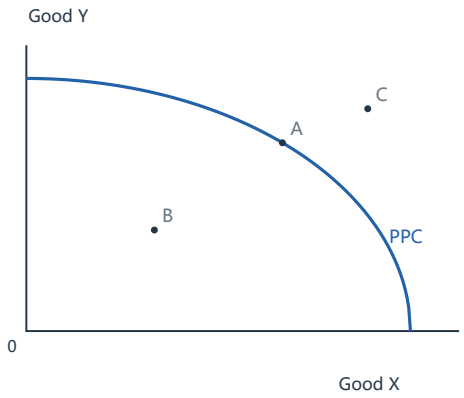
1.5 Relevant costs & benefits

Include explicit payments and implicit opportunity costs. Ignore **sunk costs**: past costs that cannot be recovered. Fixed costs do not vary with current output; they are not necessarily sunk.

1.6 Marginal analysis & consumer choice

Increase an activity while marginal benefit exceeds marginal cost. Diminishing MU means each extra unit adds less satisfaction; total utility still rises if MU > 0. A price fall creates substitution and real-income effects.

> Interior optimum: MB = MC. Consumer: MUx/Px = MUy/Py, subject to the budget. Discrete choices may not satisfy exact equality.



PPC: A is efficient; B is feasible but inefficient; C is currently unattainable.

> **REMEMBER** A change in productive capacity shifts the PPC; a change in resource use moves the point.

Demand, Supply & Elasticity

2.1 Demand

Own price changes cause movement **along** D. Income, tastes, related-good prices, expectations and buyer numbers **shift** D. Income rises: normal-good D rises; inferior-good D falls. Substitute price rises: D rises. Complement price rises: D falls.

2.2 Supply

Own price changes cause movement along S. Lower input costs, better technology, more sellers or per-unit subsidies shift S right. Higher variable costs or per-unit taxes shift S left. Market demand and supply add individual quantities **horizontally**.

2.3 Price elasticity of demand

Classify by $|E_d|$: **elastic > 1; unit elastic = 1; inelastic < 1**. More substitutes, longer adjustment time and a larger budget share usually increase elasticity. A downward-sloping linear demand curve has constant slope, **not** constant elasticity.

$$|E_d| = \left| \frac{\%\Delta Q_d}{\%\Delta P} \right| = \left| \frac{\frac{Q_2 - Q_1}{(Q_1 + Q_2)/2}}{\frac{P_2 - P_1}{(P_1 + P_2)/2}} \right|$$

> **Total revenue:** TR = P × Q. Elastic: P and TR move in opposite directions; inelastic: same direction; unit elastic: TR is unchanged.

2.4 Price elasticity of supply

Es measures producers' percentage quantity response to price. More time, spare capacity and flexible inputs generally increase Es. Vertical S: Es = 0. Horizontal S: perfectly elastic.

> Es = % change in Qs / % change in P.

2.5 Other elasticities

Cross-price elasticity: positive = substitutes; negative = complements. Income elasticity: positive = normal good; negative = inferior good. Keep the signs for these measures.

> Exy = % change in Qx / % change in Py. Ei = % change in Q / % change in income.



Left: Demand rises: D1 to D2. With supply unchanged, price and quantity both rise. **Right:** Surplus preview (2.6): CS above Pe; PS below Pe, up to Qe.

> **REMEMBER** Own price changes quantity demanded/supplied. Other determinants shift demand/supply.

Welfare & Market Policy

2.6 Equilibrium & surplus

Equilibrium: Qd = Qs. CS is below D and above the buyer price. PS is above S and below the seller price, up to quantity traded. Without market failure, competitive equilibrium maximizes total surplus.

> TS = CS + PS. Triangle area = 1/2 x base x height. PS = TR - TVC; economic profit = PS - TFC.

2.7 Disequilibrium & shifts

Below equilibrium price: shortage (Qd > Qs). Above it: surplus (Qs > Qd). D rises: P and Q rise. S rises: P falls, Q rises. With simultaneous shifts, the opposing effect requires information about relative magnitudes.

2.8 Controls, taxes & subsidies

Binding ceiling < equilibrium price: shortage. Binding floor > equilibrium price: surplus. Without government purchases, the short side limits trade. A unit tax reduces Q; the **less elastic side bears more**. A subsidy usually increases Q; without externalities it can cause overproduction.

> Tax = buyer price - seller price. Revenue = tax x Q after tax. Linear-model DWL = 1/2 x tax x reduction in Q. Subsidy spending = subsidy x Q.

2.9 Trade, tariffs & quotas

Small-country model: world price below the closed-economy price means imports; above means exports. Imports = Qd - Qs. A tariff raises domestic P, increases domestic output and reduces consumption/imports. A quota limits imports; quota rents do not automatically go to government.

> Tariff revenue = tariff x imports after tariff. DWL = production-distortion triangle + consumption-distortion triangle.



Left: Pc - Pp = tax. Qt is at D and S + tax. Yellow = tax revenue; purple = DWL. **Right:** Imports shrink from Qd0 - Qs0 to Qd1 - Qs1. Yellow = revenue; purple = DWL.

Two shifts: D up + S up: Q rises, P unclear. D up + S down: P rises, Q unclear. Reverse both directions for the opposite results.

> **REMEMBER** Ceiling below equilibrium: shortage. Floor above equilibrium: surplus. Tax burden depends on elasticity, not who legally pays.

Production, Costs & Profit

3.1 Production function

Short run: at least one fixed input. Long run: all inputs adjustable. Diminishing marginal returns: MP eventually falls as a variable input is added to fixed inputs. Falling MP does **not** imply falling total output.

> MP = change in Q / change in input. AP = Q / input. MP > AP raises AP; MP < AP lowers AP.

3.2 Short-run costs

MC crosses AVC and ATC at their minima. AFC falls as Q rises. A fixed-cost increase raises ATC and AFC, not MC or AVC. With fixed wages and labor as the only variable input, falling MP raises MC.

> TC = TFC + TVC. ATC = TC/Q = AFC + AVC. MC = change in TC / change in Q. MC = w/MP; AVC = w/AP.

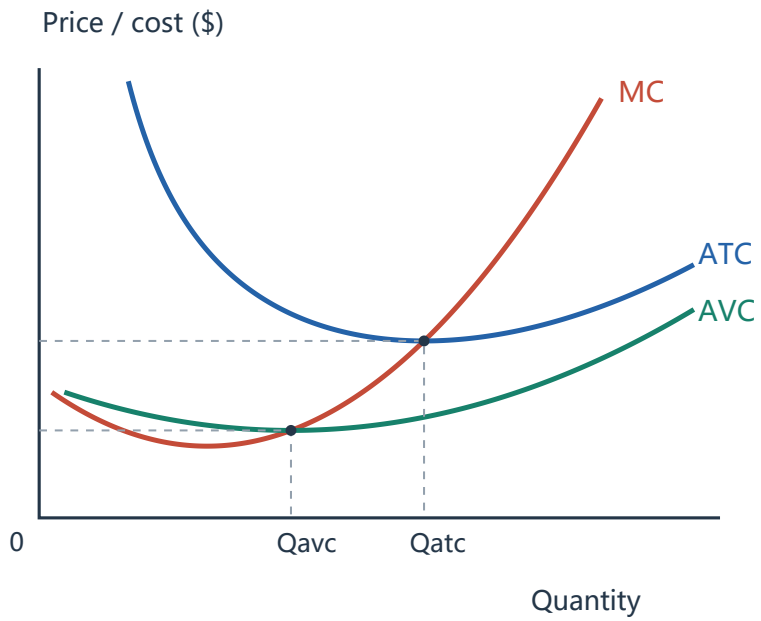
3.3 Long-run costs

LRATC is the lowest achievable average cost at each output. Economies of scale: LRATC falls; constant returns: flat; diseconomies: rises. Minimum efficient scale is the smallest output reaching minimum LRATC. Scale effects adjust **all** inputs; diminishing marginal returns holds some inputs fixed.

3.4 Profit

Accounting profit subtracts explicit costs. Economic profit subtracts explicit **and implicit** costs. Zero economic profit includes a normal return; it is not zero income.

> Economic profit = TR - TC = (P - ATC)Q. Example: P = 12, ATC = 10, Q = 5 gives profit = 10.



MC crosses AVC and ATC at their minima. ATC - AVC = AFC.

REMEMBER Zero economic profit includes the opportunity cost of the owner's resources.

Firm Decisions & Competition

3.5 Profit maximization

Increase output if MR > MC; reduce it if MR < MC. Use the rising-MC intersection. With discrete units, produce the last unit with MR at least MC; check shutdown and other feasible choices. Revenue maximization is a different objective.

> Choose MR = MC, then check whether operating is preferable to shutdown.

3.6 Shutdown versus exit

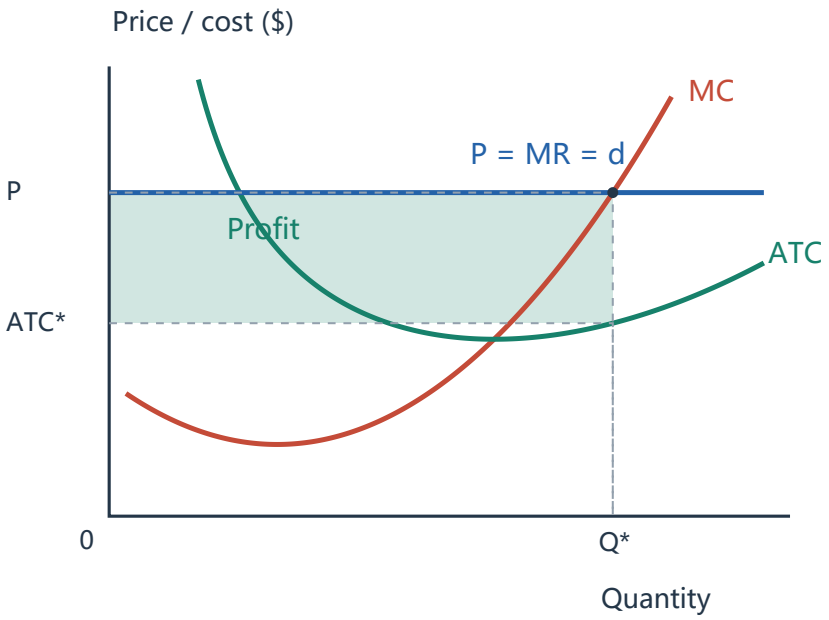
Short run: produce if revenue covers variable cost; losses can still be minimized by operating. Competitive firm: shut down if P < minimum AVC; indifferent at equality. Long run: persistent economic losses cause exit; positive profit attracts entry when permitted.

> AVC < P < ATC: operate at a loss. P = ATC: zero economic profit. P > ATC: positive economic profit.

3.7 Perfect competition

Many firms, identical products and free entry/exit. Industry S and D set price; each firm faces horizontal d = P = MR. Firm short-run supply is rising MC at/above minimum AVC. Entry shifts industry S right and reduces profit.

> Standard long-run equilibrium: P = MR = MC = minimum ATC. Productive efficiency: minimum ATC. Allocative efficiency: P = MC, absent externalities.



Q* is where MC = MR. Profit = (P - ATC*)Q*. MC crosses ATC at its minimum.

Quick check

P = 8, AVC = 6, ATC = 11, Q = 10: operating loss = 30; shutdown loss = TFC = 50. **Operate in the short run.**

> **REMEMBER** Find output using MC, then judge profit using ATC and shutdown using AVC.

Market Power & Monopoly

4.1 Market power & barriers

Imperfect competitors face downward demand. Uniform pricing makes MR < P because extra sales require price cuts on existing units. Entry barriers include patents, resource control and scale economies. Monopoly has one seller; oligopoly has a few interdependent sellers; monopolistic competition has many differentiated sellers.

4.2 Monopoly

Choose Q at MR = MC, then move **up to D** to read P. Usually P > MC and output is below the efficient level D = MC, creating DWL. Monopoly has no unique demand-independent supply curve. Entry barriers can sustain profits, but do not guarantee them.

> If P = a - bQ, then MR = a - 2bQ. Example: P = 30 - Q, MC = 6 gives Q = 12 and P = 18.

4.3 Price discrimination

Requires market power, separation of buyers and limited resale. Charge more to less price-sensitive groups. Perfect discrimination charges each unit's willingness to pay: output reaches D = MC, CS = 0 and monopoly quantity DWL disappears in the standard model. PS still excludes fixed cost.



Left: P = 30 - Q; MC = 6. Qm = 12, Pm = 18; efficient Qe = 24. Purple = DWL. **Right:** Perfect discrimination: produce to D = MC. CS = 0; shaded area = PS.

> **REMEMBER** Monopoly: MR = MC -> Q -> up to demand for P. Do not read price from MR.

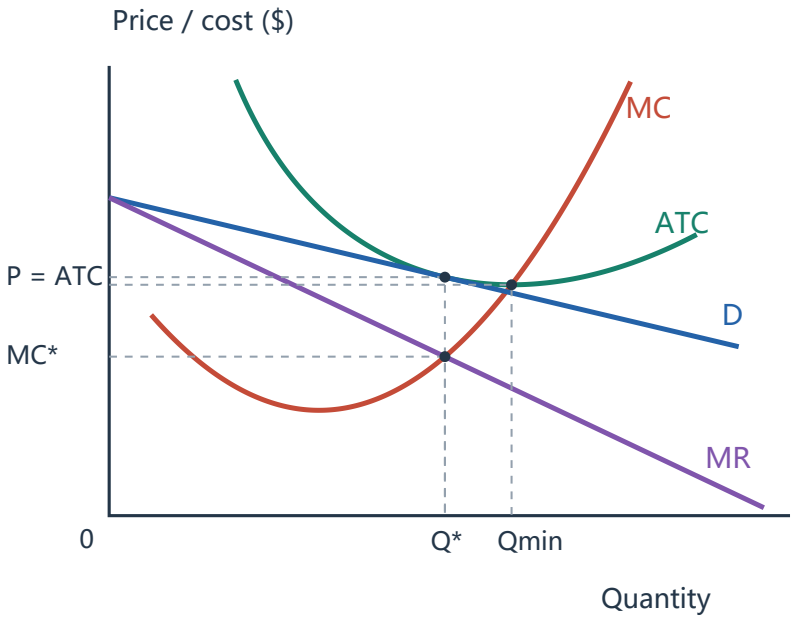
Competition & Strategy

4.4 Monopolistic competition

Differentiated products, many firms and free entry. Short run: choose MR = MC and read P from D. Entry reduces each incumbent's demand. Long run: D is tangent to ATC; P = ATC but P > MC. Output is below minimum-ATC output: **excess capacity**.

4.5 Oligopoly & game theory

A dominant strategy is best regardless of the rival's choice. A Nash equilibrium is a set of mutual best responses: nobody gains by changing alone. Collusion can raise joint profit, but cheating incentives undermine it. Nash outcomes need not maximize joint profit; a dominant strategy need not exist.



D is tangent to ATC at Q*. MR = MC at the same Q*. Excess capacity: Qmin - Q*.

Payoff matrix: (A payoff, B payoff)

A / B	B: high price	B: low price
A: high price	(8, 8)	(2, 10)
A: low price	(10, 2)	(4, 4)

Low price is dominant for each firm. Nash equilibrium: **(low, low)**, earning (4, 4). Joint profit is higher at (high, high), but either firm can gain by undercutting.

REMEMBER Zero long-run profit does not guarantee productive or allocative efficiency.

Factor Markets

5.1 Derived factor demand

Firms demand labor because it helps sell output. MP measures extra physical output; MRP measures extra revenue. Product-market competition and labor-market competition are **separate assumptions**.

> MRP = MP x MR. Only when the product market is competitive does MRP = MP x P.

5.2 Factor-market shifts

Higher product demand/price or higher productivity usually raises labor demand. More available qualified workers raises labor supply. Wage changes move along curves. Another input's price can create substitution and output effects; the net labor-demand effect depends on the situation.

5.3 Competitive factor markets

Market S and D determine wage. An individual competitive employer faces horizontal labor supply, so MFC = w. A binding minimum wage above competitive equilibrium usually lowers employment and creates excess labor supply.

> Hire where MRP = MFC. Competitive labor market: MRP = w. Cost-minimizing input mix: MPL/w = MPK/r (interior solution).

5.4 Monopsony

A powerful buyer faces upward labor supply. With a uniform wage, hiring more requires raises for existing workers, so MFC > wage. Choose L at MRP = MFC, then read wage from **labor supply**. Compared with competition, wage and employment are lower. A suitably set minimum wage can raise both.



Left: Competitive employer: hire where MRP = wage = MFC. **Right:** Monopsony: MRP = MFC sets Lm; supply sets wm.

> **REMEMBER** Monopsony: MFC sets hiring; supply sets the wage. Do not confuse monopoly with monopsony.

Efficiency & Externalities

6.1 Social efficiency

Social efficiency maximizes total social benefit minus total social cost. If MSB > MSC, increase activity; if MSB < MSC, reduce it. DWL lies between MSB and MSC over the distorted quantity range. Efficiency and distributional fairness are different judgments.

> Social optimum: MSB = MSC. Private market: MPB = MPC.

6.2 Externalities

Negative production externality: MSC > MPC, so output is too high. Positive consumption externality: MSB > MPB, so output is too low. Negative consumption: MSB < MPB. Positive production spillovers can make MSC < MPC. Standards, tradable permits or bargaining may help; bargaining needs clear rights and low transaction costs.

> Corrective unit tax = MEC at Q*. Corrective subsidy = marginal external benefit at Q*. MSC = MPC + MEC; MSB = MPB + MEB.

6.3 Public goods & common resources

Public goods are nonrival and nonexcludable: free riding causes underprovision. Add willingness to pay **vertically** for public goods. Common resources are rival but nonexcludable: overuse is the tragedy of the commons. Private goods are rival/excludable; club goods are nonrival/excludable before congestion.



Left: Negative production externality: Qmarket > Q*. Purple = DWL. **Right:** Positive consumption externality: Qmarket < Q*. Purple = lost net benefit.

> **REMEMBER** Private equilibrium: MPB = MPC. Social optimum: MSB = MSC.

Policy & Inequality

6.4 Government policy & market structure

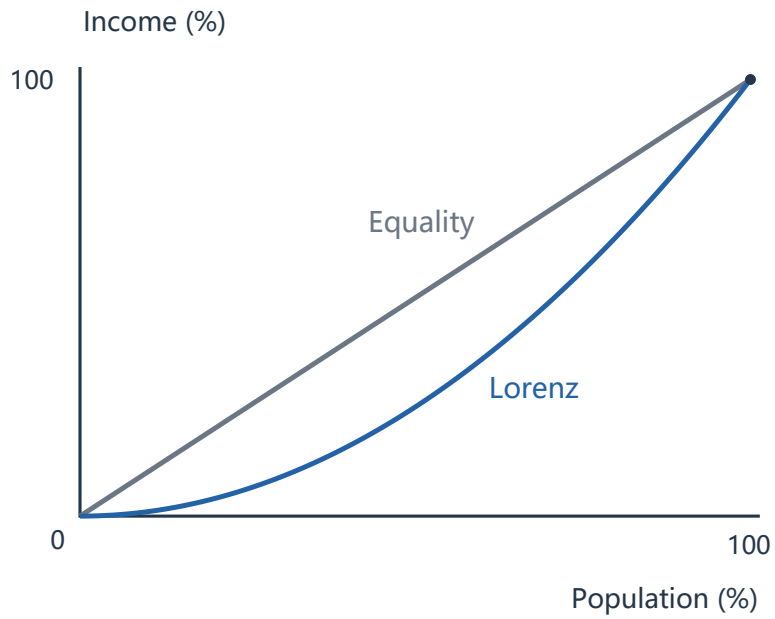
A lump-sum tax/subsidy changes profit, not MC; short-run output is unchanged if the firm keeps operating. A per-unit tax raises MC and usually lowers monopoly output. Natural monopoly: scale economies favor one supplier. Antitrust seeks to protect competition.

P = MC regulation: allocatively efficient, but may require subsidy = (ATC - P)Q. P = ATC regulation: zero economic profit, usually below efficient output.

6.5 Inequality

Income is a flow; wealth is a stock of assets minus liabilities. Lorenz curves farther below equality generally show greater inequality; higher Gini means more inequality. Sources include skills, inheritance, discrimination, access and bargaining power. Progressive/regressive/proportional taxes have rising/falling/constant average tax rates.

Average tax rate = total tax / income. Understand Lorenz/Gini comparisons; drawing Lorenz curves and calculating Gini are outside the AP requirement.



Both axes are cumulative percentages; population is ranked poorest to richest.

FRQ checklist

Draw: label axes, curves, equilibrium, prices and quantities. **Calculate:** formula, substitution, units. **Explain:** state the curve/condition, its change and the resulting outcome.

Official scope & practice

College Board: course framework, effective Fall 2026
Official FRQs and scoring guidelines
OpenStax: cost-curve relationships

All 36 topics covered. Curves rebuilt from consistent functions; graphs use independent example scales. Revised graphics: 2026-10-16.

REMEMBER Fairness and efficiency are separate. Compare average tax rates, not just dollars paid.